

GVPT Data Science Camp

Data Communication

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- ▶ Learn how to combine code, results, and figures into one document

Learning Objectives

- ▶ Learn how to combine code, results, and figures into one document
- ▶ Introduction to markdown, Quarto, Overleaf, and $\text{\LaTeX}$

The screenshot displays the Quarto editor interface. On the left, the source file 'hello.qmd' is open, showing a YAML header and R code. The code includes a title, format, editor, and library loading for 'tidyverse' and 'palmerpenguins'. Below the code, the text 'Meet Quarto' and 'Meet the penguins' is visible, along with a small illustration of three penguins labeled 'CHINSTRAN', 'GENTOO', and 'ADELIEY'. The right pane shows the rendered HTML document, which includes the title 'Hello, Quarto', a subtitle 'Meet Quarto', and a paragraph about Quarto. Below this, the text 'Meet the penguins' is followed by a paragraph about the 'palmerpenguins' data and a scatter plot titled 'Flipper and bill length' showing the relationship between flipper and bill lengths for three penguin species: Adelia, Chinstrap, and Gentoo.

```

---
title: "Hello, Quarto"
format: html
editor: visual
---

{r}
#| label: load-packages
#| include: false

library(tidyverse)
library(palmerpenguins)
  
```

Meet Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see <https://quarto.org>.

Meet the penguins

The `penguins` data from the [palmerpenguins](#) package contains size measurements for `r` `nrow(penguins)` penguins from three species observed on three islands in the Palmer Archipelago, Antarctica.

The plot below shows the relationship between flipper and bill lengths of these penguins.

```

{r}
#| label: plot-penguins
  
```

Hello, Quarto

Meet Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see <https://quarto.org>.

Meet the penguins

The `penguins` data from the [palmerpenguins](#) package contains size measurements for 344 penguins from three species observed on three islands in the Palmer Archipelago, Antarctica.

The plot below shows the relationship between flipper and bill lengths of these penguins.

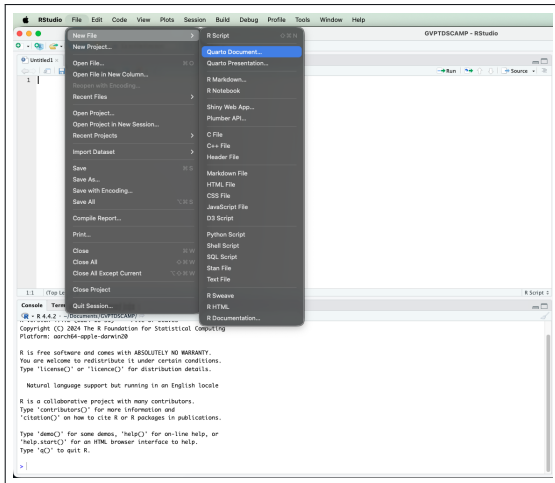
Flipper and bill length
Dimensions for penguins at Palmer Station LTER

The scatter plot shows the relationship between flipper and bill lengths for three penguin species: Adelia (orange), Chinstrap (purple), and Gentoo (teal). The x-axis represents flipper length in millimeters, and the y-axis represents bill length in millimeters. The plot shows a positive correlation between flipper and bill lengths for all three species.

Source: Quarto

- ▶ A simple and easy-to-use markup language to format your documents

Create a Quarto Document



Create a Quarto Document

New Quarto Document

 Document

 Presentation

 Interactive

Title:

Author:

☒ **HTML**
Recommended format for authoring (you can switch to PDF or Word output anytime)

☐ **PDF**
PDF output requires a LaTeX installation (e.g. <https://yihui.org/tinytex/>)

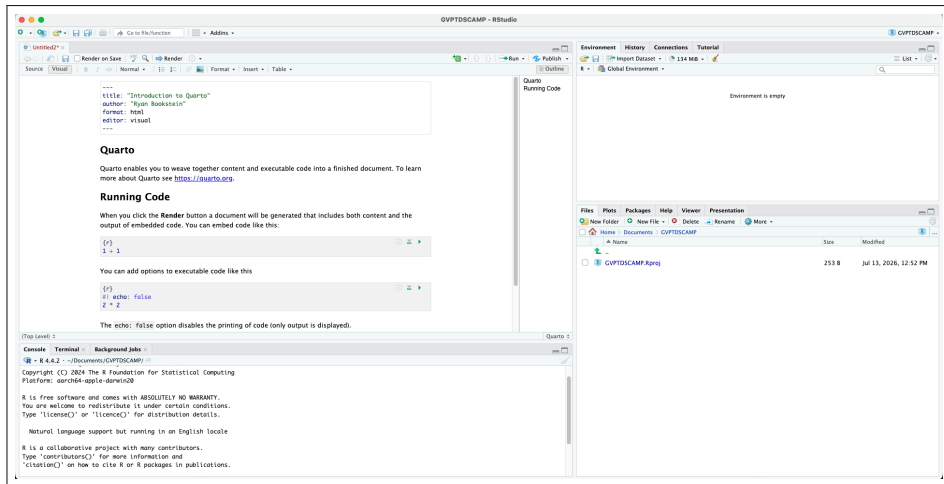
☐ **Word**
Previewing Word documents requires an installation of MS Word (or Libre/Open Office on Linux)

Engine:

Editor: ☒ Use visual markdown editor [?](#)

[? Learn more about Quarto](#)

Create a Quarto Document



Rendering your Document

The screenshot displays the RStudio interface with a Quarto document open. The editor on the left shows the source code of the document, which includes a title, author, and two code blocks. The viewer on the right shows the rendered HTML output, which includes the title, author, and the output of the code blocks. The console at the bottom shows the R environment.

```
1 ---
2 title: "Introduction to Quarto"
3 author: "Ryan Bookstein"
4 format: html
5 editor: visual
6 ---
7
8 # Quarto
9
10 Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see
11 <https://quarto.org>.
12
13 # Running Code
14
15 When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can
16 embed code like this:
17
18 ```{r}
19 1 + 1
20 ```
21
22 You can add options to executable code like this
23
24 ```{r}
25 #| echo: false
26 2 + 2
27 ```
28
29 The 'echo: false' option disables the printing of code (only output is displayed).
```

Introduction to Quarto

AUTHOR
Ryan Bookstein

Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see <https://quarto.org>.

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

[1] 2

You can add options to executable code like this

```
1 + 1
```

[1] 4

The `echo: false` option disables the printing of code (only output is displayed).

R 4.4.2 (2024-04-24) (64-bit)
Type 'license()' or 'licence()' for distribution details.
Natural language support but running in an English locale
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' for how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> |

- ▶ Code chunks start with ````{r}` and end with `````

```
```{r}
#| echo: true

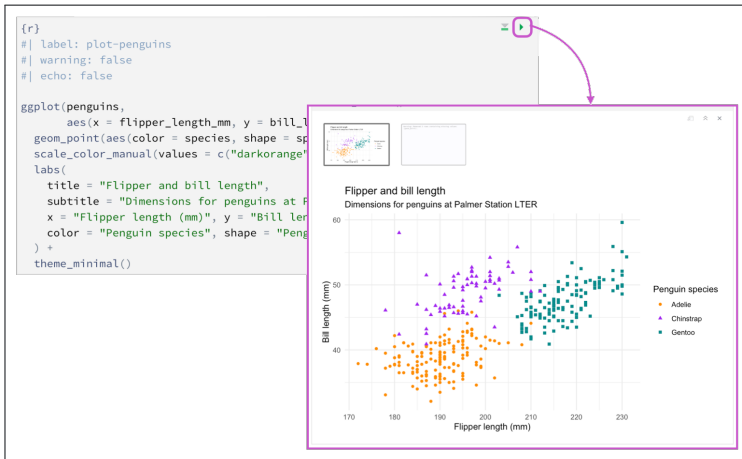
library(tidyverse)
```
```

- ▶ You can control how code chunks render and are displayed

```
```{r}  
#| echo: true
#| warning: false

library(tidyverse)
```
```

Code Chunks in Quarto



Source: Quarto

- ▶ Overleaf is an online $\text{\LaTeX}$ editor used to write, edit, and publish scientific documents

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- ▶ $\text{\LaTeX}$ is a high-quality document preparation system and markup language

Creating an Overleaf Account

- ▶ Open your browser and navigate to overleaf.com/edu/umd

Creating an Overleaf Account

- ▶ Open your browser and navigate to overleaf.com/edu/umd
- ▶ Click Log In or Register

Creating an Overleaf Account

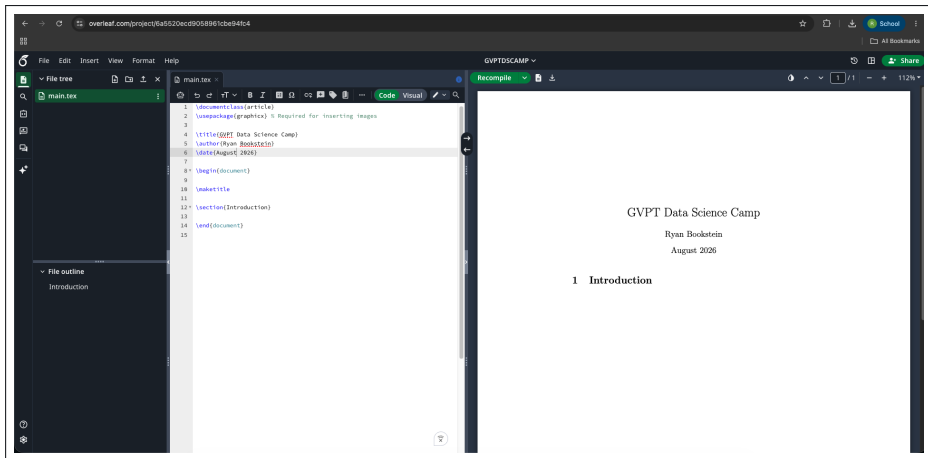
- ▶ Open your browser and navigate to overleaf.com/edu/umd
- ▶ Click Log In or Register
- ▶ Select the option to login through your institution (SSO) and use your UMD Directory ID and password to link your account

- ▶ Navigate to your project dashboard

- ▶ Navigate to your project dashboard
- ▶ Select New project, Blank project

- ▶ Navigate to your project dashboard
- ▶ Select New project, Blank project
- ▶ Name it

- ▶ Navigate to your project dashboard
- ▶ Select New project, Blank project
- ▶ Name it
- ▶ Select Create



Setting up Your Document

- ▶ Every Overleaf document requires a “preamble” at the top to define the page layout

```
\documentclass{article}
\usepackage[utf8]{inputenc}

\title{My First Document}
\author{Your Name}
\date{August 27, 2026}

\begin{document}
\maketitle

Hello World! This is my first Overleaf document.

\end{document}
```

- ▶ Unlike other word processors, you do not highlight text to format it. Wrap the text in commands to achieve desired formatting

```
\textbf{GVPT Data Science Camp.}
```

```
\textit{GVPT Data Science Camp.}
```


- ▶ Keep your document organized by using built-in sectioning commands

```
\section{Introduction}  
This is the introduction.  
  
\subsection{Background}  
Here is some background information.
```

Questions?